

MCM — Mutation Control Module

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Category: Governance & Enforcement

Module: Mutation Control Module (MCM)

Type: Control Module

Standard: Self-Evolving Systems Control Standard (SESCS)

Version: 1.0

Status: Canonical Module

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Authority: OOF™ Origin Open Foundation™

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English (UCL™)

Compatibility:

SESCS · MTVF · EVIP · ArtData · AI Governance Systems · ORGS

Canonical Definition System

Mutation Control Module (MCM) defines the structural boundaries, permissions, and constraints under which a self-evolving system is allowed to modify its behavior, logic, or execution pathways.

Base

Self-evolving systems without defined mutation boundaries:

- exceed intended operational scope
- create uncontrolled behavior
- introduce hidden risk
- cannot be reliably governed

Control must exist **before mutation occurs**.

Protocol

Any system capable of self-modification **MUST**:

- define allowed mutation zones
- define restricted mutation zones
- define prohibited mutation zones
- enforce execution permissions before mutation

Absence of defined mutation control results in:

Uncontrolled Mutation State (UMS)

Scope

MCM applies to:

- AI agents modifying internal logic
 - systems generating new executable code
 - adaptive decision systems
 - autonomous optimization engines
 - runtime self-modifying environments
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Methodology

1. Identify mutation capability
 2. define mutation scope
 3. classify mutation zones
 4. assign permission rules
 5. enforce mutation execution control
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Mutation Zones

Allowed Zone

- predefined safe mutation scope
- validated and monitored changes

→ mutation permitted

Restricted Zone

- sensitive system areas
- conditional mutation

→ mutation permitted under strict validation

Prohibited Zone

- core logic
- safety systems
- control layers

→ mutation not permitted

Structural Rule (Critical)

No mutation is permitted outside defined mutation zones.

MIF — Minimal Implementation Framework

Step 1 — Mutation Identification

Identify all system components capable of mutation.

Minimum:

- mutation points defined
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Step 2 — Zone Classification

Assign mutation zones:

- allowed
 - restricted
 - prohibited
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Step 3 — Permission Enforcement

Define mutation permissions.

Minimum:

- execution control before mutation
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Step 4 — Boundary Enforcement

Ensure mutation cannot occur outside defined zones.

Minimum:

- structural enforcement mechanism
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MIF+ — Extended Implementation Framework

Includes all MIF requirements plus:

- dynamic mutation boundary adjustment
 - risk-based mutation scoring
 - automated enforcement policies
 - multi-layer permission validation
 - integration with TSA classification
 - continuous runtime boundary monitoring
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Decision Logic

Mutation state is:

- **CONTROLLED** → within defined zones
 - **UNCONTROLLED** → outside defined zones
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Use Case 1 — AI Agent Code Generation

Scenario

AI agent generates new execution logic.

Problem

- agent modifies core system behavior
- uncontrolled code injection
- system instability

Application

MCM defines:

- allowed mutation scope
- prohibited core areas

Result

- agent operates within boundaries
- core system remains stable
- mutation risk is controlled

Use Case 2 — Autonomous Optimization System

Scenario

System continuously optimizes operational decisions.

Problem

- optimization affects critical system layers
- risk of unintended system behavior

Application

MCM enforces:

- restricted zones for sensitive logic
- conditional mutation validation

Result

- optimization remains controlled
- system integrity preserved
- risk reduced

Compatibility

MCM operates with:

- SESCO — control layer
- MTVF — validation layer
- TSA — trust classification
- EVIP — ethical governance
- ArtData — clean data inputs
- ORGS — system-level compatibility

Structural Principle

Control must exist before change.
Without boundaries, evolution becomes instability.

Closing Statement

A system that can modify itself without limits cannot be controlled.

Mutation Control Module (MCM) establishes a non-negotiable condition:

Mutation is permitted only within defined, enforced, and validated boundaries.

Related Documents

[→ SESCO Standard](#)

[→ SESCO — Master Index](#)

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